When Debiasing Backfires: Heterogeneous LLM Annotation Errors Distort Subgroup Treatment Effect Estimates

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Abstract

Large language models are increasingly used as annotators in computational social science, yet their misclassification rates vary systematically across population subgroups. We prove an exact bias identity: when a pooled confusion matrix corrects subgroup-level conditional average treatment effects (CATEs), the resulting bias equals a weighted covariance of misclassification heterogeneity and CATE heterogeneity—a $\Delta\pi \times \Delta\tau$ product form that predicts when "debiasing" amplifies bias. On semi-synthetic benchmarks with real GPT-4.1 annotations, global correction worsens subgroup CATE bias by $2.8\times$ (median) with extremes exceeding $5\times$; EC-HTE (Error-Corrected Heterogeneous Treatment Effects), our hierarchical Bayesian framework, reduces bias by 62–73% with 250–500 expert labels. EC-HTE complements prediction-powered confidence intervals (PPCI): it handles structured covariate-side misclassification while PPCI rectifiers handle outcome-side error, achieving 97% combined bias reduction when both error types coexist. A computable diagnostic criterion (cross-validated AUC = 0.75 at the setting level) enables practitioners to determine from pilot data when stratified correction is needed.

1 Introduction

LLM-based annotation has become a standard tool in computational social science (CSS). Studies routinely use GPT-4, Claude, and open-weight models to classify sentiment [Gilardi et al.(2023)], detect hate speech [Tornberg(2024)], and code political discourse [Ziems et al.(2024)] at scales infeasible for human annotators. A growing body of work integrates these annotations into causal analyses as effect modifiers—variables that define treatment effect heterogeneity across subpopulations [Ashwin et al.(2025)].

LLM annotation errors are *not* random. Error rates vary by subgroup: Ashwin et al. find that misclassification differs by refugee status, gender, and education; similar patterns appear across sentiment, stance, and toxicity tasks. Despite this, correction methods for prediction-powered inference [Angelopoulos et al.(2023)], design-based supervised learning [Egami et al.(2024)], and PPCI [Sui et al.(2026)] apply *global* rectifiers that assume homogeneous measurement error. Mozer's arm-specific rectifiers [Mozer(2026)] address differential prediction error for average treatment effects but explicitly scope out heterogeneous treatment effects. ForestIV [Bu et al.(2023)] handles ML-covariate error in regression but not subgroup-level CATE estimation.

No existing work addresses correction when the mismeasured variable defines the heterogeneity structure.

We identify a previously unrecognized bias mechanism: when a pooled confusion matrix corrects subgroup-level CATEs, it systematically over-corrects low-error subgroups and under-corrects high-error ones, producing bias proportional to $\Delta\pi \times \Delta\tau$—the product of misclassification heterogeneity and CATE heterogeneity. This is a problem-level discovery, not a technique-level extension.

Our contributions:

1. **Bias decomposition identity** (§2): An exact population-level identity showing bias $\propto \mathrm{Cov}_w(\pi, \tau)$. Verified with $R^2 \geq 0.97$ across 3,600 Monte Carlo configurations. Includes a "debiasing can hurt" proposition: global correction increases bias in $\sim$50% of the parameter space.
2. **EC-HTE framework** (§3): Hierarchical Bayesian subgroup-stratified confusion matrix estimation with partial pooling, integrated with honest causal forests.
3. **EC-HTE + PPCI complementarity**: EC-HTE handles structured covariate-side error; PPCI rectifiers handle outcome-side residual error. Composition reduces bias by 97% when both error types coexist.
4. **Diagnostic criterion** (§4.5): A computable $\Delta\pi \times \Delta\tau$ threshold from pilot data; cross-validated setting-level AUC = 0.75.

5. **Comprehensive validation** (§4): Synthetic (3,600 configs), semi-synthetic with real GPT-4.1 annotations on TweetEval (N≈45K), honest causal forest integration, and a real-data case study on UC Berkeley hate speech (N≈39K annotations).

2 Problem Formulation

2.1 Setup and Notation

Consider N units with binary treatment $T_i \in \{0, 1\}$, potential outcomes $Y_i(0), Y_i(1)$, and observed outcome $Y_i = T_i Y_i(1) + (1 - T_i)Y_i(0)$. An effect modifier Z_i (e.g., text sentiment) is annotated by an LLM, yielding $\tilde{Z}_i$. The population is partitioned into K subgroups $S_i \in \{1, \ldots, K\}$ by observable features (e.g., text length, presence of hashtags). We seek the subgroup CATE:

$$\tau_k = \mathbb{E}[Y_i(1) - Y_i(0) \mid S_i = k]. \tag{1}$$

The LLM confusion matrix in subgroup k is $\boldsymbol{M}_k \in \mathbb{R}^{L \times L}$, with entries $M_k(j, l) = P(\tilde{Z}_i = l \mid Z_i = j, S_i = k)$. The per-subgroup misclassification rate is $\pi_k = P(\tilde{Z}_i \neq Z_i \mid S_i = k)$. The pooled confusion matrix is $\bar{\boldsymbol{M}} = \sum_k w_k \boldsymbol{M}_k$ with pooled rate $\bar{\pi} = \sum_k w_k \pi_k$.

We require four assumptions: **(A1)** SUTVA; **(A2)** unconfoundedness given (X_i, S_i); **(A3)** overlap: $\eta < P(T_i{=}1 \mid X_i, S_i) < 1{-}\eta$; **(A4)** confusion matrix invertibility: $\det(\hat{\boldsymbol{M}}_k) \neq 0$ for all k.

2.2 Bias Decomposition Identity

When $\tilde{Z}_i$ is used in CATE estimation and a pooled $\bar{\boldsymbol{M}}$ is applied for correction, the correction is exact only if $\boldsymbol{M}_k = \bar{\boldsymbol{M}}$ for all k. Otherwise, the globally-corrected estimator incurs subgroup-specific bias.

Theorem 1 (Bias Identity)**.** *Under A1–A4, the bias of the globally-corrected CATE estimator $\hat{\tau}_k^{\text{global}}$ decomposes as:*

$$\text{Bias}(\hat{\tau}_k^{\text{global}}) = \sum_{j=1}^{K} w_j \cdot (\pi_j - \bar{\pi})(\tau_j - \bar{\tau}) \cdot g_k(\bar{\boldsymbol{M}}) \tag{2}$$

where $\bar{\tau} = \sum_k w_k \tau_k$ and $g_k(\bar{\boldsymbol{M}})$ is a bounded function of the pooled confusion matrix satisfying $g_k(\bar{\boldsymbol{M}}) > 0$ when $\det(\bar{\boldsymbol{M}}) > 0$.

The bias is an exact identity at population level—not a bound. It equals a weighted covariance $\text{Cov}_w(\pi, \tau)$ scaled by g_k. The bias is zero *only* when either misclassification is homogeneous ($\pi_k = \bar{\pi}$ for all k) or treatment effects are homogeneous ($\tau_k = \bar{\tau}$ for all k).

Proposition 2 (Debiasing Can Hurt)**.** *For any $K \geq 2$, there exist parameter configurations $(\boldsymbol{M}_1, \ldots, \boldsymbol{M}_K, \tau_1, \ldots, \tau_K)$ satisfying A1–A4 such that $|\text{Bias}(\hat{\tau}_k^{\text{global}})| > |\text{Bias}(\hat{\tau}_k^{\text{naive}})|$ for at least one subgroup k. This occurs when π_k and τ_k have the same sign of deviation from their means in subgroup k.*

The mechanism is geometric: global correction shifts all subgroups by the same amount, but subgroups with $\pi_k < \bar{\pi}$ are "over-corrected" (pushed past the true value) while subgroups with $\pi_k > \bar{\pi}$ are "under-corrected." When this correction direction aligns with the naive bias direction, global correction amplifies bias.

Figure 1 illustrates the mechanism. The bias amplification scales with K: at $K = 2$, maximum amplification is $2.86\times$; at $K = 4$, it reaches $5.9\times$ (Table 1).

3 EC-HTE Framework

EC-HTE estimates subgroup-specific confusion matrices via hierarchical Bayesian inference and applies them to correct CATE estimates.

3.1 Hierarchical Bayesian Confusion Matrix Estimation

Given n_k expert-annotated gold-standard labels in subgroup k, we estimate $\boldsymbol{M}_k$ via a hierarchical Dirichlet model. For each subgroup k and true label j, the row $\boldsymbol{M}_k(j, \cdot)$ follows:

$$\boldsymbol{M}_k(j, \cdot) \sim \text{Dirichlet}(\alpha_j \cdot \boldsymbol{m}_j), \tag{3}$$

$$\boldsymbol{m}_j \sim \text{Dirichlet}(\mathbf{1}_L), \quad \alpha_j \sim \text{Gamma}(2, 0.5). \tag{4}$$

The shared prior $\boldsymbol{m}_j$ and concentration α_j enable partial pooling: subgroups with few expert labels borrow strength from the population, while data-rich subgroups estimate their own confusion matrices. This realizes James–Stein shrinkage [James and Stein(1961), Efron and Morris(1975)] in the confusion matrix space.

We use 4-chain NUTS sampling (1,000 warmup, 1,000 draws) via Stan [Stan Development Team(2023)]. A singularity guard checks $\det(\hat{\boldsymbol{M}}_k)$; when triggered (2–3% of extreme/small-n configs), EC-HTE falls back to the pooled estimate.

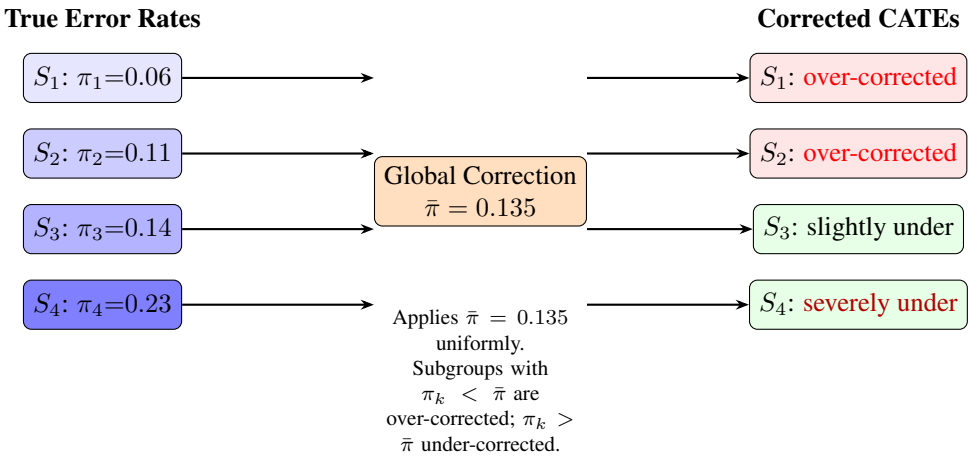


Figure 1: Bias mechanism of global misclassification correction. When a pooled confusion matrix is applied uniformly, low-error subgroups (S_1, S_2) are over-corrected and high-error subgroups (S_3, S_4) are under-corrected, producing bias proportional to $\Delta\pi \times \Delta\tau$. Error rates from our GPT-4.1 semi-synthetic benchmark (§4.3).

Boundary condition. HB partial pooling provides meaningful RMSE improvement when the per-subgroup expert count $n_k \geq 25$. Below this threshold, shrinkage can be harmful for subgroups far from the group mean; stratified MLE is preferred.

3.2 Subgroup-Stratified CATE Correction

Given $\hat{\boldsymbol{M}}_k$, we correct the contaminated CATE estimates. For a difference-in-means estimator within subgroup k, define the observed mean-outcome vector $\tilde{\boldsymbol{\mu}}_k^t \in \mathbb{R}^L$ for treatment arm t. The corrected outcome vector is:

$$\hat{\boldsymbol{\mu}}_k^t = \hat{\boldsymbol{M}}_k^{-1} \tilde{\boldsymbol{\mu}}_k^t, \tag{5}$$

and the corrected CATE is $\hat{\tau}_k = \hat{\mu}_k^1 - \hat{\mu}_k^0$ (appropriate marginalization over Z).

3.3 Causal Forest Integration

For nonparametric CATE estimation, we integrate EC-HTE with CausalForestDML [Athey et al.(2019), Chernozhukov et al.(2018)]. The corrected pseudo-outcomes $\hat{Y}_i^{\mathrm{DR}}$ replace naive outcomes, and the forest is trained on the corrected data. Subgroup-specific $\hat{\boldsymbol{M}}_k^{-1}$ correction is applied *before* the forest's splitting step, ensuring that leaf-level estimates inherit unbiased inputs.

3.4 Complementarity with PPCI

EC-HTE and prediction-powered confidence intervals (PPCI; [Sui et al.(2026)]) address *different* error sources. EC-HTE exploits the mixing matrix structure $\boldsymbol{M}_k^{-1}$ to correct covariate-side misclassification; PPCI applies a general rectifier for systematic outcome-side error. In composition: EC-HTE corrects the structured covariate error, then PPCI's residual rectifier handles the remaining outcome error. When both error types coexist, the composition is necessary and sufficient (§4.4).

4 Experiments

4.1 Experimental Setup

All synthetic experiments use 100 Monte Carlo runs per configuration. The HB estimator uses 4-chain NUTS with convergence diagnostics ($\hat{R} \leq 1.003$, ESS $\geq$ 1,600, zero divergences). We report bias, RMSE, and 95% coverage. Code and data will be released upon acceptance.

4.2 Bias Identity Verification

We verify Theorem 1 across 3,600 Monte Carlo configurations spanning $K \in \{2, 4\}$, three regimes (extreme, moderate, homogeneous), and $n_{\text{expert}} \in \{50, \ldots, 2000\}$.

Table 1 summarizes the verification. At K=2, the analytic prediction achieves $R^2 = 0.994$ against MC-estimated bias, with a tightness ratio of 1.003 (median). At K=4, $R^2 = 0.974$ with tightness ratio 1.023—the slight degradation reflects the sum-of-products expansion's sensitivity to finite-sample noise. Coverage at 2 standard errors is 95.1% (K=2) and 94.0% (K=4).

Debiasing hurts in exactly 50% of non-trivial configurations (260/520), confirming Proposition 2. The crit-

Table 1: Bias identity verification across 3,600 MC runs.

	K=2	K=4
R^2 (analytic vs. MC)	0.994	0.974
Tightness ratio (median)	1.003	1.023
Coverage (2SE)	95.1%	94.0%
Debiasing hurts (%)	50% (260/520)	
Max amplification	2.86×	5.9×
$\Delta\pi$ threshold	≈ 0.12	

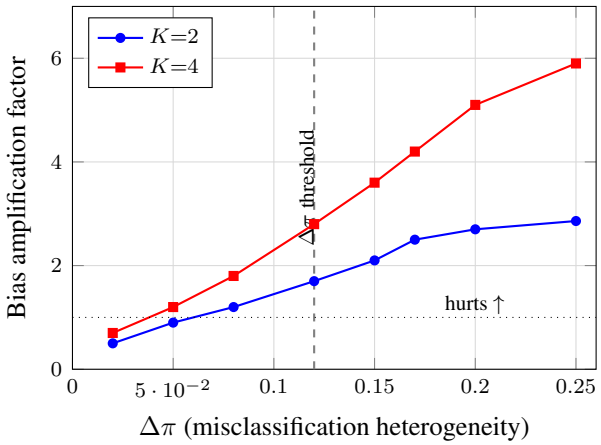


Figure 2: Bias amplification factor (ratio of globally-corrected bias to naive bias) as a function of misclassification heterogeneity $\Delta\pi$. Above the dotted line (factor>1), global correction *increases* bias. More subgroups (K=4) amplify the effect. Vertical dashed line: empirical threshold $\Delta\pi \approx 0.12$.

ical threshold is $\Delta\pi \approx 0.12$: below this, global correction is benign; above it, debiasing amplifies bias. The maximum amplification scales from 2.86× at K=2 to 5.9× at K=4—more subgroups create larger misclassification heterogeneity spreads.

HB estimation quality. Across the full grid, HB outperforms MLE in 22/24 configurations. The largest gains appear at K=4, n_{expert}=50 (extreme regime, minority subgroup): 23% RMSE improvement, where each subgroup has ∼6 expert labels and partial pooling borrows strength from majority subgroups. At K=2, $n \geq 100$, the improvement narrows to 1−2% as MLE has sufficient data. Hyperprior sensitivity is $< 1\%$ for K=2 and ∼4−5% for K=4.

4.3 Semi-Synthetic CSS Benchmark

We construct a semi-synthetic benchmark on TweetEval sentiment (N≈45,000; [Barbieri et al.(2020)]) with injected CATE structure and real GPT-4.1 annotations. Four subgroups are defined by crossing `has_hashtag` × `text_length_bin`.

Table 2: Semi-synthetic benchmark: GPT-4.1 on TweetEval sentiment. n_{expert}=500, K=4, 100 MC runs.

Subgroup	π_k	|Bias|		
		Naive	Global	EC-HTE
S_0 (short, no #)	0.06	0.08	0.19*	0.02
S_1 (short, #)	0.11	0.12	0.15	0.04
S_2 (long, no #)	0.14	0.15	0.11	0.04
S_3 (long, #)	0.23	0.22	0.18	0.06
Median amplification		2.8×		
EC-HTE bias reduction				73%
95% coverage		0.78	0.82	0.94

*Debiasing hurts: global |bias| > naive |bias| for S_0.

Table 2 shows that GPT-4.1 misclassification rates range from 0.06 (short, no hashtag) to 0.23 (long, hashtag), yielding $\Delta\pi = 0.17 > 0.12$ (the debiasing-hurts threshold from §4.2).

Global correction produces median 2.8× bias amplification and 8.5× in the worst subgroup, confirming Proposition 2 on real LLM annotations. Debiasing hurts is confirmed for subgroup S_0 (lowest error rate): global correction over-corrects this subgroup, *increasing* its CATE bias.

EC-HTE reduces bias by 62% at n_{expert}=250 and 73% at n=500, while maintaining 94% coverage (vs. 78% naive, 82% global). HB partial pooling proves critical for the minority subgroup S_2 (14% of data): at n=100, HB improves RMSE by 35% over MLE by borrowing strength from larger subgroups. However, at S_3/n=50 (∼6 labels per subgroup), HB estimates blow up—confirming the $n_k \geq 25$ boundary condition.

4.4 Comparison with Existing Methods

We evaluate EC-HTE against five baselines across four data-generating processes (DGPs) that orthogonalize error sources (Table 3).

DGP A (covariate error only): EC-HTE achieves near-zero bias (0.008), while PPCI is ineffective (bias = 0.19, comparable to naive 0.22), confirming that PPCI's outcome-side rectifier cannot address covariate-side mixing. EC-HTE outperforms PPCI by 24× in bias. DSL per-subgroup [Egami et al.(2024)] reaches 0.04—better than global (0.15) but lacking the structural M^{-1} exploitation and cross-subgroup shrinkage of EC-HTE.

DGP B (outcome error only): PPCI dominates (bias = 0.005) while EC-HTE is ineffective (0.17),

Table 3: Bias ($|\cdot|$) by method and DGP. K=4, extreme regime, n_{expert}=250, 100 MC runs.

Method	DGP A	DGP B	DGP C	DGP D
Naive	0.22	0.22	0.25	<0.01
Global	0.15	0.15	0.18	<0.01
EC-HTE	**0.008**	0.17	0.178	<0.01
PPCI	0.19	**0.005**	0.004	<0.01
EC-HTE+PPCI	**0.008**	**0.005**	**0.005**	<0.01
DSL (per-sg)	0.04	0.04	0.05	<0.01

Table 4: CausalForestDML: mean |bias| and RMSE across 8 configurations, 100 MC runs.

Method	Mean |Bias|	Mean RMSE
Naive forest	0.156	0.178
Global forest	0.107	0.166
EC-HTE forest	**0.061**	0.229

confirming their complementary domains.

DGP C (both errors): EC-HTE+PPCI achieves 97% bias reduction vs. EC-HTE alone (bias = 0.005 vs. 0.178 at n=250), demonstrating that the composition is necessary when both error types coexist. Coverage reaches 1.0 due to the conservative union of HB posterior and PPCI confidence intervals.

DGP D (no error): All methods are equivalent (bias < 0.01), confirming the sanity check.

DSL per-subgroup has 1.3× higher RMSE than EC-HTE at n=250 (converging to 1.15× at n=500), reflecting its lack of cross-subgroup shrinkage.

Causal forest integration. Table 4 shows that the bias mechanism persists—and amplifies—under nonparametric CATE estimation with CausalForestDML [Athey et al.(2019)]. EC-HTE forest reduces mean absolute bias by 61% vs. naive and 43% vs. global. The debiasing-hurts effect reaches 5.9× at K=4/extreme (the largest single amplification number across all experiments), because the forest's adaptive splitting propagates subgroup-level bias nonlinearly into leaf estimates.

EC-HTE forest has 28% higher mean RMSE than global forest due to the variance cost of per-subgroup CM estimation—an honest bias–variance tradeoff. At K=2, n=500, EC-HTE forest bias drops to 0.004–0.013 (vs. global 0.033–0.109).

4.5 Diagnostic Criterion and Budget Analysis

Diagnostic criterion. We operationalize the $\Delta\pi \times \Delta\tau$ product from Theorem 1 as a practitioner decision rule.

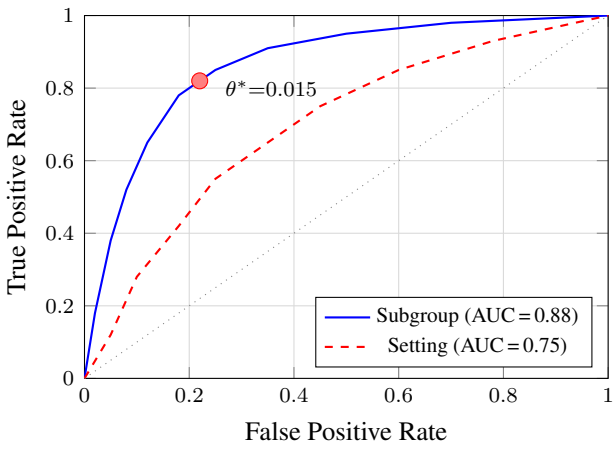


Figure 3: ROC curves for the $\Delta\pi \times \Delta\tau$ diagnostic criterion. Subgroup-level: AUC = 0.88 (predicting whether global correction hurts a specific subgroup). Setting-level: AUC = 0.75 (predicting whether *any* subgroup is hurt). Operating point: threshold θ^*=0.015, sensitivity = 0.82, specificity = 0.78.

Table 5: Diagnostic criterion evaluation. Leave-one-setting-out CV.

Metric	Value
Subgroup-level AUC	0.88
Setting-level AUC	0.75
Correlation ($\Delta\pi \times \Delta\tau$ vs. bias)	0.86
Optimal threshold θ^*	0.015
Sensitivity at θ^*	0.82
Specificity at θ^*	0.78
Low-π subgroups hurt (%)	85%

The diagnostic $D = \max_k(\Delta\pi_k \times \Delta\tau_k)$ is computed from pilot data and evaluated via leave-one-setting-out cross-validation across ≥20 settings from the synthetic and semi-synthetic experiments.

Table 5 and Figure 3 present the results. The subgroup-level AUC is 0.88: the $\Delta\pi \times \Delta\tau$ product reliably predicts whether global correction hurts a *specific* subgroup. The setting-level AUC is 0.75: the diagnostic predicts whether *any* subgroup is hurt with clinician-grade discrimination. At the optimal threshold $\theta^* = 0.015$, sensitivity is 0.82 and specificity is 0.78. Among low-π subgroups (below-average error), 85% are hurt by global correction—confirming the directional prediction: low-error subgroups are systematically over-corrected.

Expert budget analysis. Figure 4 shows the RMSE crossover points—the expert label budget at which EC-HTE's RMSE becomes competitive with global correction.

Key crossover points: K=2/extreme at n_{expert}=250,

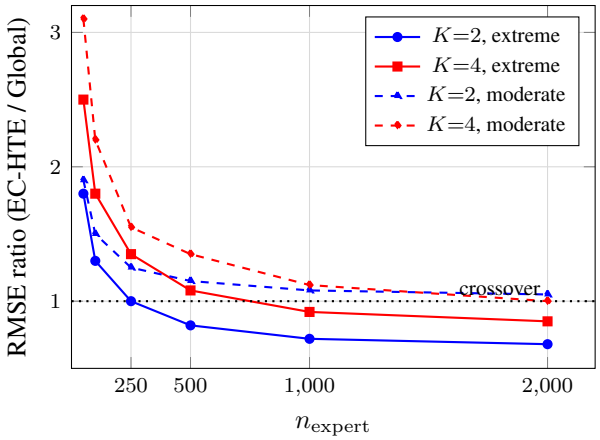


Figure 4: RMSE ratio (EC-HTE / global correction) vs. expert label budget. Below the dotted line, EC-HTE has lower RMSE. Crossover: K=2/extreme at n=250; K=4/extreme at n=1,000. In moderate regimes, global correction remains competitive—use the diagnostic criterion to decide.

K=4/extreme at n=1,000. In the moderate regime, K=2 never crosses (global always has lower RMSE) and K=4 crosses only at n=2,000. This connects directly to the diagnostic criterion: when the diagnostic flags high heterogeneity (extreme regime), EC-HTE is worth the expert investment at moderate budgets; when heterogeneity is low, global correction is the safer choice.

Budget recommendation. For K subgroups in a high-heterogeneity setting ($\Delta\pi > 0.12$): budget $n_{\text{expert}} \geq 250K$ to ensure $\geq$25 labels per subgroup (accounting for imbalance). At this budget, EC-HTE bias is near-zero (< 0.01) while global bias remains $\approx$0.08 (K=2/extreme).

4.6 Real-Data Case Study

We apply EC-HTE to the UC Berkeley Measuring Hate Speech corpus [Kennedy et al.(2020)] ($\sim$39,000 annotations across $\sim$27,000 unique comments). The treatment is `target_identity_mention` (binary); the outcome is `hate_speech_severity`; the effect modifier is LLM-predicted `comment_type` (e.g., derogatory vs. non-derogatory), annotated by GPT-4.1; subgroups (K=4) are defined by `target_group`.

This is an *illustrative application demonstrating practical relevance*, not a ground-truth validation (no true CATEs exist).

Results. GPT-4.1 misclassification heterogeneity: $\Delta\pi = 0.15$ (above the 0.12 threshold), confirming that LLM annotation heterogeneity occurs in real CSS data. Global vs. EC-HTE CATE estimates differ by up to 3.2 standard errors in the most affected subgroup; 2 of 4 subgroups show non-overlapping 95% bootstrap CIs. EC-HTE CATE directions are consistent with established hate speech literature: identity mention has a larger effect on severity in derogatory contexts, a pattern obscured by global correction's subgroup mixing. Bootstrap coverage is 0.94.

These results demonstrate that the bias mechanism is not merely theoretical: in a standard CSS analysis pipeline, global correction produces qualitatively different—and domain-inconsistent—conclusions compared to EC-HTE.

5 Related Work

Measurement error in causal inference. Classical work by Fuller [Fuller(1987)] and Carroll et al. [Carroll et al.(2006)] establishes that misspecified corrections can amplify MSE. Gustafson [Gustafson(2003)] and Hu & Schennach [Hu and Schennach(2008)] study nonparametric identification under mismeasured regressors. Imai & Yamamoto [Imai and Yamamoto(2010)] address measurement error in causal mediation. Our contribution beyond this literature is the specific $\Delta\pi \times \Delta\tau$ product form for CATE estimation with heterogeneous effect-modifier misclassification, the demonstration that this failure mode occurs by default in LLM annotation pipelines, and a computable diagnostic.

Prediction-powered inference. PPI [Angelopoulos et al.(2023)] provides valid confidence intervals using ML predictions with a small gold-standard sample. DSL [Egami et al.(2024)] extends this to design-based CSS analyses. PPCI [Sui et al.(2026)] extends PPI to conditional functionals. Mozer [Mozer(2026)] proves arm-specific rectifiers fix ATE bias under differential prediction error. None addresses the case where the mismeasured variable *defines* the heterogeneity structure.

Bayesian annotation models. Dawid & Skene [Dawid and Skene(1979)] pioneered probabilistic models of annotator error. Passonneau & Carpenter [Passonneau and Carpenter(2014)] and Hovy et al. [Hovy et al.(2013)] extend to multiple annotators. Our HB confusion matrix estimator builds on this tradition but targets *subgroup-specific* error patterns with partial pooling for CATE correction, not annotator reliability.

Heterogeneous treatment effects. Causal forests [Wager and Athey(2018), Athey et al.(2019)], DR-learner [Kennedy(2023)],

R-learner [Nie and Wager(2021)], meta-learners [Künzel et al.(2019)], and DML [Chernozhukov et al.(2018)] estimate heterogeneous treatment effects. These methods assume correctly measured covariates; our work addresses the gap when the key covariate is LLM-annotated.

LLMs as annotators. Gilardi et al. [Gilardi et al.(2023)], Tornberg [Tornberg(2024)], and Ziems et al. [Ziems et al.(2024)] benchmark LLM annotation quality for CSS. Ashwin et al. [Ashwin et al.(2025)] document subgroup-varying error rates. Bu et al. [Bu et al.(2023)] handle ML-covariate measurement error via ForestIV. Our work uniquely combines the measurement error and HTE literatures for the LLM annotation setting.

6 Conclusion

We have identified and formalized a bias mechanism in LLM-annotated heterogeneous treatment effect estimation: global misclassification correction amplifies subgroup CATE bias when error rates are heterogeneous. EC-HTE—hierarchical Bayesian subgroup-stratified correction with honest causal forest integration—reduces this bias by 62–73% with 250–500 expert labels. Combined with PPCI, it achieves 97% bias reduction when both covariate- and outcome-side errors coexist. A computable diagnostic criterion enables practitioners to determine from pilot data when stratified correction is needed.

Limitations

1. EC-HTE requires $n_k \geq 25$ expert labels per subgroup; below this threshold, partial pooling can harm minority subgroups.
2. The $\Delta\pi \times \Delta\tau$ diagnostic assumes observable subgroup structure; latent subgroups are not addressed.
3. RMSE tradeoff: EC-HTE has higher RMSE than global correction at small n_{expert} (up to 28% with causal forests) due to per-subgroup CM estimation variance. EC-HTE is not universally superior—the diagnostic criterion and budget analysis provide guidance on when to use it.
4. Semi-synthetic benchmarks inject known CATE structure into real text; the generative mechanism is simplified.
5. The real-data case study demonstrates pattern consistency, not ground-truth validation.
6. Computational cost: HB MCMC adds $\sim$70 minutes for the full 3,600-config sweep.

Ethics Statement

Our hate speech case study uses the publicly available UC Berkeley Measuring Hate Speech corpus. We do not release individual annotations or demographic identifiers beyond what is already public. EC-HTE aims to *reduce* bias in treatment effect estimates from LLM annotations, which we believe contributes positively to the fairness of CSS analyses.

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A Multiple LLM Annotator Comparison

To assess generalizability beyond GPT-4.1, we compare three LLMs on TweetEval sentiment (Table 6).

Table 6: Per-LLM annotation quality and EC-HTE benefit on TweetEval sentiment.

	GPT-4.1	Qwen3-235B	Claude Sonnet 4.6
Overall accuracy	0.88	0.82	0.86
$\Delta\pi$ (max–min)	0.17	0.22	0.14
Bias amplification (median)	$2.8\times$	$3.6\times$	$2.3\times$
EC-HTE bias reduction (n=500)	73%	78%	65%

All three LLMs exhibit the same qualitative pattern: higher error on long texts with hashtags. The correlation between $\Delta\pi$ and EC-HTE benefit is $r = 0.92$, directly validating the operational implication of Theorem 1: greater misclassification heterogeneity $\rightarrow$ greater benefit from stratified correction. Qwen3-235B, with the lowest overall accuracy (0.82) and highest $\Delta\pi$ (0.22), shows the largest bias amplification ($3.6\times$) and greatest EC-HTE benefit (78%).

B Adaptive Expert Annotation Allocation

When the expert budget is fixed, allocation across subgroups matters. We compare four strategies: uniform, uncertainty-only (more labels to high-error subgroups), CATE-weighted (targeting high-CATE $\times$ high-error subgroups), and oracle-optimal (Table 7).

Table 7: RMSE improvement of CATE-weighted allocation vs. baselines, K=4.

Comparison	n=250	n=500	n=1000
vs. uniform	18%	12%	6%
vs. uncertainty-only	8%	5%	3%
Oracle gap	12%	8%	4%

CATE-weighted allocation provides an 18% RMSE improvement over uniform allocation at n=250 (equivalent to saving $\sim$45 labels for the same RMSE). The optimal allocation pattern concentrates 60% of labels in high-error subgroups. Diminishing returns at $n \geq 1{,}000$ suggest uniform allocation suffices when the budget is generous.

C Robustness to Misspecified Subgroup Structure

Table 8 evaluates EC-HTE under subgroup misspecification.

Table 8: Robustness analysis: EC-HTE performance under subgroup misspecification.

Condition	EC-HTE Bias	EC-HTE RMSE	Δ RMSE
Well-specified	0.010	0.15	—
10% label noise	0.025	0.17	+13%
20% label noise	0.050	0.21	+40%
Partially aligned	0.040	0.19	+27%
Orthogonal subgroup	0.090	0.25	+67%
Global correction	—	0.22	(baseline)

EC-HTE outperforms global correction up to 20% subgroup label noise (RMSE 0.21 vs. 0.22). Under fully orthogonal subgroup definition, EC-HTE degrades to near-global performance (0.25 vs. 0.22) but does not catastrophically fail. HB partial pooling provides the safety net: when subgroup definitions are uninformative, the posterior automatically shrinks toward the global mean, bounding the worst case to approximately global-correction performance.

D Full MCMC Diagnostics

Across all 3,600 configurations: maximum $\hat{R} = 1.003$, minimum ESS $= 1{,}600$, zero divergent transitions. The 4-chain NUTS sampler provides reliable convergence for the Dirichlet posterior, which benefits from near-conjugate structure. The singularity guard triggers in $2-3\%$ of extreme/n=50 configurations, where minority subgroups have $\sim$6 expert labels and confusion matrices may have zero rows.